

```
hist(iris$Sepal.Width)
mean(iris$Sepal.Width)
[1] 3.057333
median(iris$Sepal.Width)
[1] 3
quantile(iris$Sepal.Width, 0.73)
73%
3.3
pairs(iris[,1:4])
> pairs(iris[,1:4])
> data(PlantGrowth)
> PlantGrowth
weight group
1 4.17 ctrl
2 5.58 ctrl
3 5.18 ctrl
4 6.11 ctrl
5 4.50 ctrl
6 4.61 ctrl
7 5.17 ctrl
8 4.53 ctrl
9 5.33 ctrl
10 5.14 ctrl
11 4.81 trt1
12 4.17 trt1
13 4.41 trt1
```

14 3.59 trt1

15 5.87 trt1

16 3.83 trt1

17 6.03 trt1

18 4.89 trt1

19 4.32 trt1

20 4.69 trt1

21 6.31 trt2

22 5.12 trt2

23 5.54 trt2

24 5.50 trt2

25 5.37 trt2

26 5.29 trt2

27 4.92 trt2

28 6.15 trt2

29 5.80 trt2

30 5.26 trt2

```
> data(PlantGrowth)
```

```
>
```

```
> hist(
```

```
+ PlantGrowth$weight,
```

```
+ breaks = seq(3.3, max(PlantGrowth$weight) + 0.3, by = 0.3),
```

```
+ main = "Histogram of Plant Weights",
```

```
+ xlab = "Weight",
```

```
+ ylab = "Frequency"
```

```
+ )
```

```

> save.image("C:\\Users\\josea\\Desktop\\R stuff\\Week 3 assignment.R")

> > # Minimum weight in trt2

> min_trt2 <- min(PlantGrowth$weight[PlantGrowth$group == "trt2"])

>

> # Percentage of trt1 weights below that minimum

> mean(PlantGrowth$weight[PlantGrowth$group == "trt1"] < min_trt2) * 100

[1] 80

> > # Keep only plants with weight > 5.5

> heavy_plants <- subset(PlantGrowth, weight > 5.5)

>

> # Count observations in each group

> group_counts <- table(heavy_plants$group)

>

> # Create a colorful barplot

> barplot(
+ group_counts,
+ col = rainbow(length(group_counts)),
+ main = "Plants with Weight > 5.5 by Group",
+ xlab = "Group",
+ ylab = "Count"
+ )

```